Enhancing Pedagogical Feedback Quality of Large Language Models via Learning Analytics in a Kanban-Based Platform for Student Self-Regulated Learning Support

Anonymized for Review

***Abstract*—While visualization tools like Kanban boards facilitate process monitoring, they lack the capacity for adaptive guidance. To address this limitation and the scalability crisis of human expert feedback, this study aims to enhance the quality of pedagogical feedback generated by Large Language Models (LLMs) to support student Self-Regulated Learning (SRL). The enhancement focuses on three core functions: feedback, motivation, and appreciation. A comparative experimental method was employed without fine-tuning, evaluating various Context Engineering schemes enriched with Learning Analytics (LA) on two models: Llama 3.1 8B and Llama 3.3 70B. Evaluation was quantitative (BERTScore, BARTScore, LLM-as-a-Judge) and qualitatively validated by educational psychology experts.**

Results indicate that the effectiveness of the enhancement significantly depends on the model size. The smaller model (8B) achieved the highest performance increase with complex context (ReAct + LA), suggesting a reliance on rich, structured guidance to compensate for its internal limitations. Conversely, the larger model (70B), despite having a superior Baseline performance, showed degraded performance when given overly complex context. The best-performing model (Llama 8B with ReAct + LA) was validated by human experts as "Usable with minor revision". The study concludes that LA-enriched context engineering is an effective, practical, and computationally efficient strategy for improving pedagogical feedback, particularly for smaller LLMs.

***Index Terms*—Large Language Model, Self-Regulated Learning, Learning Analytics, Context Engineering, Pedagogical Feedback**

I. Introduction

Self-Regulated Learning (SRL)—encompassing Zimmerman's cyclical phases of forethought, performance, and self-reflection—is strongly linked to academic achievement [1], [2], yet many students lack the skills to apply it effectively [3], [4]. Personalized expert feedback is the most effective scaffold for SRL development [5], [6], but faces a scalability crisis as rising class sizes limit lecturer–student interaction [7], [8].

Large Language Models (LLMs) offer a scalable complement, yet their feedback tends to be generic, overly generous, and process-agnostic rather than guiding self-regulation [9], [10]. This study addresses these limitations through *context engineering*—the systematic enrichment of LLM prompts with structured information—without model fine-tuning. Specifically, *Learning Analytics* (LA) signals derived from a Kanban-based platform (card movements, completion timestamps, stuck indicators) are injected into the prompt to enable personalized, data-driven pedagogical feedback across three dimensions: **feedback**, **motivational support**, and **appreciative support** [5], [11].

To guide this investigation, three research questions are addressed:

1) **RQ1:** To what extent can context engineering combined with Learning Analytics improve LLM pedagogical feedback quality compared to a structured Baseline prompt?
2) **RQ2:** How do human experts perceive the quality, relevance, and usability of feedback generated by the best-performing context engineering scheme?
3) **RQ3:** To what extent does model size (8B vs. 70B) influence the effectiveness of context engineering for pedagogical feedback generation?

The main contributions are: (1) a comparative evaluation of four LA-enriched context engineering schemes on two open-source LLMs of contrasting scales; and (2) an empirical finding that model size fundamentally determines optimal context complexity, validated through automated metrics (BERTScore, BARTScore, LLM-as-a-Judge) and expert assessment.

II. Related Work

A. LLM Feedback Alignment via Fine-Tuning

Alignment training—RLHF [12] and Direct Preference Optimization (DPO) [13]—has been shown to improve LLM feedback quality beyond what parameter scaling achieves: InstructGPT (1.3B, RLHF) outperforms GPT-3 (175B), and a DPO-tuned Llama 3.1 8B surpasses GPT-4o in preference and tone positivity in educational grading tasks. However, both approaches require substantial labeled data and computational resources that limit applicability in resource-constrained academic settings.

B. Prompt Engineering and Learning Analytics

Without retraining, prompt quality is the primary lever for LLM feedback improvement: Jacobsen and Weber [14] show that only high-quality prompts reliably produce outputs that surpass novice teacher feedback, though affective dimensions remain inconsistent. Enriching prompts with learning analytics further advances this direction—Li and Ma [15] demonstrate that LA-informed context engines (LACE, KCE) yield curriculum-aligned SRL feedback in programming education, while Strickroth et al. [11] confirm that Kanban process traces are both pedagogically meaningful and practically extractable (teacher SUS 92). Existing work, however, evaluates prompts in isolation or within a single domain and model scale, leaving open the question of how context complexity and model size jointly determine feedback quality across multiple SRL-relevant dimensions.

III. Methodology

A. Research Design

This study employs a **comparative experimental design** to evaluate four prompt variation conditions across two open-source LLMs—**Llama 3.1 8B Instant** and **Llama 3.3 70B Versatile**—without model fine-tuning, yielding eight experimental configurations applied to 40 synthetic student datasets. The system was implemented in Python 3.12 with `LangChain` and `LangGraph` for prompt orchestration, inference via the Groq API, and **GPT-4o** (OpenAI API) as the LLM-as-a-Judge evaluator. Evaluation combines automated quantitative metrics with qualitative expert validation, as illustrated in Fig. 1.

B. Theoretical Framework

To formally characterize the role of context engineering in LLM inference, this study adopts the lens of implicit Bayesian inference [16]. This formulation provides a principled explanation of how enriching the prompt—with SRL rubrics and learning analytics signals—shapes the model's internal belief about the student's learning state, thereby directing the generated feedback toward pedagogically appropriate responses.

1) Variable Definitions: Let M denote the LLM used for inference. The input to M consists of two components:

1) **Context Engineering** $C = (c_1, c_2, c_3, c_4, c_5)$, comprising five components: role and persona definition (c_1), internal logic and assessment rubric (c_2), SRL phase mapping (c_3), assessment prompt (c_4), and output format specification (c_5).
2) **Student Data** $D = (K, A)$, where $K = \{\kappa_1, \kappa_2, \ldots, \kappa_m\}$ is the set of m Kanban board cards with attributes $\kappa_j = (\mathit{title}_j, \mathit{status}_j, t_j^{\text{created}}, t_j^{\text{moved}}, \mathit{label}_j)$, and $A = \varphi(K)$ is the learning analytics vector derived from K through a transformation function φ that computes descriptive indicators such as completion rate, stuck rate, and velocity.

The complete prompt provided to the model is the concatenation:

$$\mathit{Prompt} = [C,\ D] = [c_1, c_2, c_3, c_4, c_5,\ K,\ \varphi(K)] \quad (1)$$

2) Inference Formulation: The model M generates a response $y = (y_1, y_2, \ldots, y_T)$ as a sequence of T tokens. At each step, the model implicitly marginalizes over a set of latent concepts Θ learned during pretraining, where each $\theta \in \Theta$ represents a student learning pattern (e.g., on-track, struggling, procrastinating):

$$p_M(y_t \mid C, D, y_{<t}) = \int_{\theta \in \Theta} p(y_t \mid \theta, C, y_{<t}) \cdot p(\theta \mid C, D, y_{<t})\, d\theta \quad (2)$$

Here, $p(\theta \mid C, D, y_{<t})$ is the model's implicit posterior belief about the student's learning pattern. The context engineering C forms a prior expectation by defining assessment criteria and SRL phases, while the student data D provides observational evidence for updating that belief. The final response is obtained by:

$$y^* = \arg\max_y \ p_M(y \mid C, D) \quad (3)$$

This response y^* constitutes the pedagogical feedback covering three functions—feedback, motivation, and appreciation—which is evaluated as described in Section III-E.

C. Dataset Construction and Model Selection

Synthetic data was used due to the absence of public Kanban-based SRL datasets. Each of the 40 datasets represents a student's Kanban board in JSON format, derived from the Gamatutor platform schema and covering the OOP domain, generated across five behavioral profiles: Lazy (5), Slightly Lazy (10), Average (10), Diligent (10), and Very Diligent (5). Ground truth y_{ref} was established by three university lecturers with expertise in educational psychology, who independently provided structured feedback across the three target dimensions. Two open-source LLMs were selected for evaluation—**Llama 3.1 8B Instant** (USD 0.05/1M tokens, 840 tokens/s) and **Llama 3.3 70B Versatile** (USD 0.59/1M tokens, 394 tokens/s) via the Groq API [17]—to contrast a small, cost-efficient model against a large, high-capacity counterpart under identical context conditions [18], [19].

D. Context Engineering and Prompt Variations

The feedback improvement approach rests on three interdependent pillars: context engineering, learning analytics integration, and systematic prompt variation experiments.

The first pillar is the design of a theoretically grounded **Baseline prompt**, structured into five logical components—Role Definition, Internal Logic, SRL Phase Mapping, Assessment Prompt, and Input/Output Format—to fulfill specific pedagogical objectives, including adaptive guidance and self-efficacy support. Prompt quality was validated using the theory-driven prompt manual of Jacobsen and Weber [14], where this study's prompt scored 13 out of 16, classifying it as high quality. The Baseline prompt structure is illustrated in Fig. 1.

The second pillar is the integration of **descriptive learning analytics** to inject situational awareness into the model. A transformation function φ processes raw JSON Kanban data

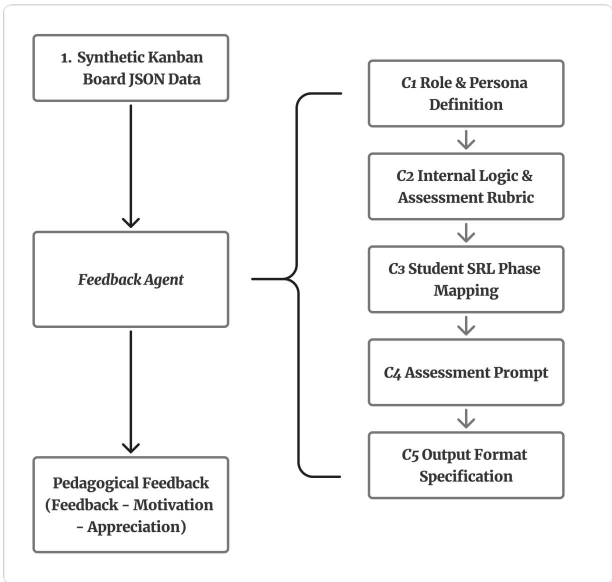

Fig. 1: Baseline prompt structure and its five context engineering components $C = (c_1, c_2, c_3, c_4, c_5)$.

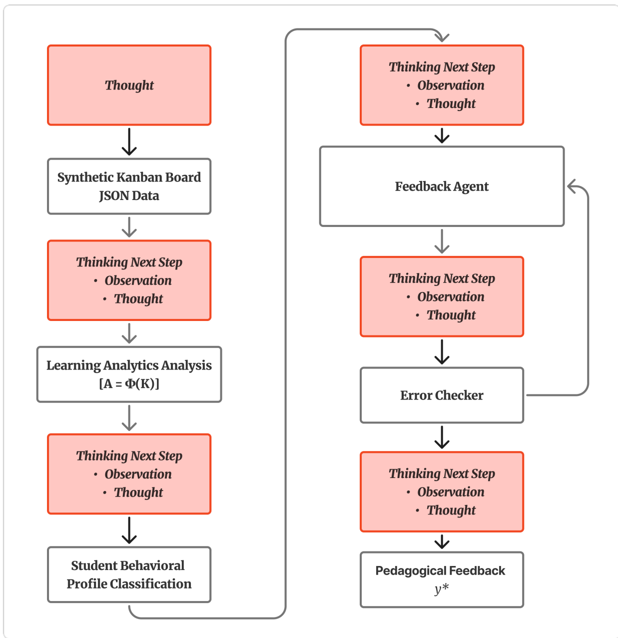

Fig. 2: ReAct agent mechanism structure.

into 12 descriptive indicators—including `aging_wip_h`, `stuck`, and `alerts`—calibrated with a 2-hour threshold appropriate for the study's context. The resulting analytic summary is appended to the prompt, enabling the model to generate personalized, data-driven feedback.

The third pillar is a systematic experiment across four **prompt variation conditions** to isolate the contribution of each contextual addition: (1) **Baseline**—the structured prompt alone; (2) **Material List**—Baseline augmented with a list of course materials; (3) **LA**—Baseline enriched with the descriptive analytics summary; and (4) **ReAct + LA**—LA-enriched prompt operating within a ReAct (Reasoning + Acting) agent implemented in LangGraph. The ReAct agent structures inference through a *Thought–Action–Observation* loop with an internal Error Checker to minimize formatting errors, as shown in Fig. 2. All four conditions were applied to both models, yielding eight experimental configurations.

E. Evaluation Framework

Evaluation compares y^* (Eq. (3)) against expert ground truth y_{ref} through three automated quantitative metrics and qualitative expert validation.

1) BERTScore: BERTScore [20] uses contextual BERT embeddings to compute token-level Precision (P_{BERT}), Recall (R_{BERT}), and F1 between y^* and y_{ref}, capturing semantic similarity beyond lexical overlap:

$$F_{\text{BERT}} = 2\,\frac{P_{\text{BERT}} \cdot R_{\text{BERT}}}{P_{\text{BERT}} + R_{\text{BERT}}} \tag{4}$$

2) BARTScore: BARTScore [21] evaluates y^* against y_{ref} by computing the weighted log-likelihood of y^* under a pretrained BART model conditioned on y_{ref}, assessing fluency and semantic coverage without retraining. Let ϕ denote the BART model parameters:

$$\text{BARTScore}(y^* \mid y_{\text{ref}}) = \sum_{t=1}^{|y^*|} w_t \log p_\phi(y_t^* \mid y_{<t}^*, y_{\text{ref}}) \tag{5}$$

where $w_t{=}1$ for uniform averaging. A less negative score indicates better semantic coverage. Both directions are computed and aggregated as Precision, Recall, and F1.

3) LLM-as-a-Judge: GPT-4o [22] acts as an automated evaluator, assigning binary quality labels per pedagogical dimension by comparing y^* against y_{ref}, achieving moderate-to-high agreement with expert human raters.

For each data point $i \in \{1, \ldots, N{=}40\}$ and dimension $d \in \mathcal{D}$, the judge returns a binary label $e_d^{(i)} \in \{0,1\}$ by comparing $y^{*(i)}$ against $y_{\text{ref}}^{(i)}$ using rubric $\mathcal{I}_d$ and analytics context $A^{(i)}$. Since expert labels are $\hat{e}_d^{(i)} = 1$ by construction, per-dimension Recall simplifies to the mean binary score; Precision and F1 are computed accordingly. The reported scores are macro-averaged over all three dimensions $\mathcal{D}$.

The prompt supplies four fields corresponding to Eqs. (1)–(3): dimension definitions $\mathcal{I}_d$, the analytics summary $A^{(i)}$, the expert reference $y_{\text{ref}}^{(i)}$, and the candidate response $y^{*(i)}$.

4) Expert Validation: An independent educational psychology expert assessed the best-performing model's outputs using a 15-indicator Likert-scale questionnaire grounded in Self-Determination Theory [23], Positive Psychology [24], and the Hattie–Timperley feedback model [25], covering the feedback, motivation, and appreciation dimensions. Statistical significance testing was not conducted; results are treated as exploratory findings.

IV. Results and Discussion

This section presents the experimental results and analysis across three evaluation dimensions—semantic alignment, pedagogical relevance, and qualitative expert assessment—and discusses the findings in relation to the three research questions introduced in Section I.

A. Semantic Alignment: BERTScore and BARTScore

Table I reports the BERTScore and BARTScore F1 scores for all eight experimental configurations. For the 8B model, both metrics indicate a consistent and monotonic improvement as context complexity increases: the ReAct + LA variant achieves the highest BERTScore F1 ($F_{\text{BERT}} = 0.6663$) and BARTScore F1 (BARTScore $= -11.40$), representing gains of $+0.0055$ and $+0.32$ respectively over the Baseline. For the 70B model, the trend is reversed: the Material List variant achieves the best performance on both metrics ($F_{\text{BERT}} = 0.6710$; BARTScore $= -11.61$), while the ReAct + LA variant yields no consistent improvement over the Baseline.

TABLE I: Quantitative Evaluation Results (F1 Scores) per Variant

Variant	Model	BERTScore F1	BARTScore F1	LLM-as-a-Judge F1
Baseline	8B	0.6608	−11.72	0.5371
	70B	0.6687	−11.69	0.7271
Material List	8B	0.6597	−11.85	0.5627
	70B	**0.6710**	**−11.61**	**0.7571**
LA	8B	0.6657	−11.54	0.5896
	70B	0.6648	−11.69	0.6399
ReAct + LA	8B	**0.6663**	**−11.40**	**0.6121**
	70B	0.6658	−11.67	0.6715

While the absolute magnitude of these improvements is modest, this is consistent with the nature of BERTScore and BARTScore: both metrics measure token-level semantic proximity to the reference y_{ref}, and expert-written feedback exhibits idiomatic variation that limits the ceiling of embedding-based similarity regardless of output quality. The more informative signal therefore lies in the consistency of the direction of improvement, particularly the steady gain observed across all three metrics for the 8B model under increasing context complexity.

B. Pedagogical Relevance: LLM-as-a-Judge

Table I also reports the LLM-as-a-Judge F1 for all variants. This metric reveals a much larger performance differential than the embedding-based metrics, particularly for the 8B model. The Baseline 8B achieves an F1 of only 0.5371, indicating that without structured context the model satisfies fewer than half of the pedagogical criteria on average. The ReAct + LA variant raises this to 0.6121, a gain of $+0.0750$—more than ten times the magnitude of the corresponding BERTScore gain. For the 70B model, the Baseline already achieves F1 = 0.7271, and the best variant (Material List, F1 = 0.7571) adds only a modest $+0.0300$.

The consistently high Precision values (1.0000 for most configurations) reflect the binary scoring design described in Section III-E: when the model satisfies a criterion, it virtually always does so correctly relative to the expert reference. The primary driver of F1 variance is therefore Recall—the proportion of expert-defined criteria actually addressed—which increases substantially for the 8B model under richer context (from 0.4567 to 0.5217) but remains near its ceiling for the 70B model (0.6417 to 0.6600 at best). This divergence suggests that the 70B model's superior pretraining already endows it with broad pedagogical criterion coverage, leaving limited headroom for context-driven improvement.

C. Effect of Model Size on Baseline Capacity

Table II summarizes the Baseline F1 scores across all three metrics. The 70B model outperforms the 8B model on every measure, with the most pronounced gap appearing in LLM-as-a-Judge F1 (0.7271 vs. 0.5371, a difference of 0.1900). In contrast, the BERTScore gap is narrow (0.6687 vs. 0.6608, $\Delta = 0.0079$), suggesting that at the surface-level vocabulary of the generated feedback, both models produce reasonably similar text. The substantially larger LLM-as-a-Judge gap confirms that the 70B model's advantage lies primarily in its capacity for pedagogically structured and criterion-satisfying response generation—a quality not captured by embedding-based metrics alone.

TABLE II: Baseline Performance Comparison: 8B vs. 70B

Metric (F1)	8B	70B	Δ (70B−8B)
BERTScore	0.6608	0.6687	+0.0079
BARTScore	-11.7282	-11.6905	+0.0377
LLM-as-a-Judge	0.5371	0.7271	+0.1900

Table I summarizes the full quantitative comparison across all variants and metrics, revealing a divergent response pattern between the two model sizes under increasing context complexity.

D. Qualitative Expert Validation

The best-performing configuration—Llama 3.1 8B with ReAct + LA—was submitted for qualitative review by an educational psychology expert independent of the three lecturers who provided y_{ref}. The expert assessed 15 indicators across three pedagogical functions using a five-point Likert scale (Table III), and rendered an overall verdict of **"Usable with minor revision."**

The Feedback dimension (mean = 3.80) is strongest in clarity, constructiveness, and feed-forward orientation (F1, F4, F5 = 4), while relevance to the specific learning context (F3 = 3) is the primary weakness, consistent with the model's limited ability to reference specific Kanban card content in its analysis. The Motivation dimension (mean = 4.00) shows the highest overall scores, with particularly strong goal-orientation (M2 = 5) aligned with Self-Determination Theory's emphasis on goal salience [23]. However, the lower relatedness score (M5 = 3) indicates a gap in empathetic connection, a finding consistent with prior work showing that affective dimensions of feedback remain difficult for LLMs to produce reliably [14]. The Appreciation dimension (mean = 3.80) demonstrates

TABLE III: Expert Qualitative Assessment (Llama 3.1 8B, ReAct + LA)

Dim.	ID	Criterion	Score
Feedback	F1	Clarity	4
	F2	Specificity	4
	F3	Relevance	3
	F4	Constructiveness	4
	F5	Feed-forward	4
		Mean	**3.80**
Motivation	M1	Intrinsic drive	4
	M2	Goal orientation	5
	M3	Autonomy	4
	M4	Competence	4
	M5	Relatedness (empathy)	3
		Mean	**4.00**
Appreciation	A1	Effort recognition	5
	A2	Proportionality	3
	A3	Confidence	4
	A4	Emotional support	4
	A5	Positive reinforcement	3
		Mean	**3.80**
		Overall Mean	**3.87**

strong effort recognition (A1 = 5), grounded in Positive Psychology's emphasis on acknowledging process over outcome [24], but scores for sincerity and positive reinforcement (A2, A5 = 3) suggest the language is perceived as formulaic rather than genuinely warm—a characteristic the expert attributed to the repetitive phrasing patterns of the 8B model noted earlier.

E. Discussion

The results collectively address the three research questions as follows.

RQ1 (effect of context engineering with LA): The integration of learning analytics consistently improves feedback quality for the 8B model across all three metrics, with the ReAct + LA variant achieving the highest quantitative scores. This confirms that LA-enriched context engineering is an effective, fine-tuning-free strategy for improving pedagogical feedback quality. The improvement pattern is consistent with the Bayesian interpretation in Eq. (2): richer context C and student data D sharpen the model's posterior belief $p(\theta \mid C, D, y_{<t})$ about the student's learning pattern, directing generation toward more criterion-satisfying outputs.

RQ2 (expert perception): The overall verdict of "Usable with minor revision" validates the practical feasibility of the approach as a complementary scaffolding system. The average expert scores across all three dimensions ($\bar{s} = 3.87/5.00$) indicate adequate pedagogical quality, with the primary revision targets being contextual specificity (F3), empathetic tone (M5), and language naturalness (A2, A5). These improvements do not require architectural changes and can be addressed through targeted prompt refinement.

RQ3 (model size): Model size fundamentally determines the optimal context complexity. The 8B model benefits monotonically from richer context—consistent with the hypothesis that smaller models compensate for limited internal capacity by relying on explicit structured guidance [18], [19]. Conversely, the 70B model achieves its best performance with minimal additional context (Material List), and its performance degrades under the ReAct + LA framework. A plausible explanation is that the highly prescriptive ReAct loop over-constrains the 70B model's generation distribution, forcing it to mimic the step-by-step, descriptive style characteristic of the 8B model rather than leveraging its superior language generation capacity. This also accounts for the qualitative observation that 70B generates more linguistically varied feedback at Baseline, while 8B tends to be repetitive and fails to clearly separate appreciation from critique.

A cross-cutting finding is the divergence between metric sensitivities: BERTScore improvements are small (max $\Delta F1 = 0.0055$) while LLM-as-a-Judge improvements are substantially larger (max $\Delta F1 = 0.0750$ for 8B). This suggests that pedagogical criterion satisfaction is more sensitive to context engineering than surface-level semantic similarity, and that LLM-as-a-Judge is a more discriminative metric for this task. Finally, the finding that a prompt scoring 13/16 on the theory-driven quality manual [14] does not guarantee optimal output—especially for the 70B model—demonstrates that prompt quality scores are a necessary but not sufficient condition for performance, and must be interpreted alongside model capacity and context complexity.

V. Conclusion

This study investigated context engineering enriched with learning analytics (LA) as a fine-tuning-free strategy for improving the pedagogical feedback quality of open-source LLMs in a Kanban-based self-regulated learning platform. Four prompt variants—Baseline, Material List, LA, and ReAct + LA—were evaluated on two models of contrasting scales (Llama 3.1 8B and Llama 3.3 70B) using a tri-metric framework combining BERTScore, BARTScore, LLM-as-a-Judge, and qualitative expert validation.

Regarding **RQ1**, the integration of learning analytics into the prompt context consistently improved feedback quality for the 8B model across all three quantitative metrics. The ReAct + LA variant achieved the highest BERTScore F1 ($F_{\text{BERT}} = 0.6663$, Δ+0.0055 over Baseline) and BARTScore F1 (BARTScore $= -11.40$, Δ+0.32), while LLM-as-a-Judge showed the largest improvement ($\Delta F1 = 0.0750$). The wider gap observed in LLM-as-a-Judge relative to embedding-based metrics confirms that pedagogical criterion satisfaction is substantially more sensitive to context enrichment than surface-level semantic similarity—underscoring LLM-as-a-Judge as the more discriminative evaluation instrument for this task.

Regarding **RQ3**, model size fundamentally determines the optimal context complexity, and larger does not imply more context-receptive. The 70B model achieved its best quantitative performance with the lightweight Material List variant ($F_{\text{BERT}} = 0.6710$), while the ReAct + LA framework produced no consistent improvement. A plausible explanation is that the prescriptive Thought–Action–Observation loop over-constrains the 70B model's generation distribution, suppressing the linguistically rich and varied output the model otherwise produces at Baseline. The 8B model, by contrast, compensates for limited internal capacity by leveraging structured guidance—consistent with the theoretical interpretation that richer context C sharpens the model's posterior belief $p(\theta \mid C, D, y_{<t})$ in Eq. (2).

Regarding **RQ2**, qualitative expert evaluation of the best-performing configuration (8B ReAct + LA) returned an aggregate verdict of "*Usable with minor revision*" ($\bar{s} = 3.87/5.00$), confirming the practical feasibility of the approach as a complementary scaffolding tool. Feedback quality was assessed positively across three pedagogical dimensions, with targeted revision recommendations centered on contextual specificity (F3), empathetic tone (M5), and language naturalness (A2, A5)— all of which are addressable through prompt refinement rather than architectural modification. Taken together, these findings establish context engineering enriched with learning analytics as an effective, resource-efficient, and practically viable strategy for improving LLM pedagogical feedback quality in scalable educational settings.

Several directions emerge from these findings. Understanding why the 70B model degrades under the ReAct + LA framework warrants further investigation, with preference optimization (DPO) or alternative agentic strategies as candidate solutions. Future studies should replicate these experiments on real student Kanban interaction logs and expand the expert panel to enable statistically robust inter-rater agreement analysis. Prompt personalization for tone and empathy—particularly for the indicators flagged by experts (M5, A2, A5)—and evaluation of models with stronger target-language proficiency remain open avenues for improvement.

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